Youri Chancrin

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Education

Engineering Degree in Computer Science — 2022 — 2025

Enseirb-Matmeca Bordeaux INP — Bordeaux, France

Specialization in High Performance Computing and Data Science

Employment

Visiting Research Intern — April 2025 — September 2025

Laboratory of Fluid Mechanics, Meiji University — Tokyo, Japan

- Reproduced and improved 3D, low-$Re \sim 200$ channel flow state prediction using GAN, achieving up to **10%** accuracy improvement in far-from-wall regions, using input scale filtering and discriminator tuning.
- Built an end-to-end data pipeline covering dataset generation with *Incompact3d*, preprocessing, object storage via *MinIO*, metadata management using *SQLAlchemy*, and model training in *PyTorch*.
- Participated actively in lab activities: delivered **5** presentations in English, **1** in Japanese, and attended academic research presentations at Shibaura Institute of Technology, Toyosu Campus.

Software Engineering Intern — June 2024 — September 2024

Capgemini Applied Innovation Exchange — Bordeaux, France

- Developed an XR application for HoloLens 2 to visualize electronic device resource costs using *Unity* and *C#*; implemented data persistence with *MongoDB Realm* and visualizations with custom *HLSL* shaders.
- Collaborated in a small Agile team; received positive feedback on documentation and usability after on-site deployment.

Projects

Article Implementation — PINN Neural Operators — *Python, PyTorch, CUDA* — 2026

- Reproduced and benchmarked a Physics-Informed Neural Operator from literature.

Incompressible Lid-Driven Cavity Solver — *C, Python, Cython* — 2026

- 2D Navier–Stokes FD solver (projection method, IMEX / ADI), stability and convergence analysis.
- Validated results within X% of reference data (Ghia et al., 1982) up to $Re = 3200$.
- Conducted comparative performance study between optimized high-level Python (NumPy, SciPy, Cython) and low-level C; achieved $\mathbf{Y}\times$ speedup through memory layout and loop-level optimization.

GEMM Optimization — *C, OpenMP, BLAS* — 2026

- Cache-aware matrix multiplication with vectorization and parallelization.
- Reached X% performance of locally compiled OpenBLAS.

Skills

Programming: Python, C, CUDA, SQL, Assembly x86, C++, C#, Java, Kotlin
Scientific & ML: NumPy, Pandas, Scikit-learn, PyTorch, Seaborn, Sympy, OpenMP, MPI
Software Engineering: Git, Linux, CMake, Docker, Hadoop DFS, Spark, Slurm
Languages: French (Native), English (C1, IELTS 7.5), Japanese (Basic)